

EEG Data Processing and Analysis Pipeline

Laboratory EEG arm

1 Inputs and recording match



XDF recording

32 channel EEG · markers



Experiment log

source clock events

Recording match

participant ↔ experiment ↔ XDF

CHECK

2 Marker recovery and event timing

Match log and EEG events

Recover event timeline

Event timing
reliable?

NO

NO · repair or
limit analysis

YES · aligned

3 EEG cleaning

Import EEG

Clean signal

Remove artifacts

EEG usable?

NO

NO · exclude with
recorded reason

YES · continue

4 Create EEG measurements

Condition windows

Compute EEG measures

Ad timing
reliable?

NO · use condition
windows only

YES

Before and after
ad windows

5 Analysis outputs

EEG feature table

Statistical comparisons

Results and figures